Thiyaneshwar

Tiruchengode | sthiyaneshwar94@gmail.com | 8438314612 |
https://www.linkedin.com/in/thiyaneshwar-s-14a429269/ | https://github.com/thiyaneshwars
https://leetcode.com/u/thiyanesh_sellappan| https://www.hackerrank.com/profile/sthiyaneshwar94

Education

Vani Vidyalaya Metric.Hr.Sec.School, Uppupalayam 2016 – 2022

- X percentage : 87%
- XII percentage : 84%

K.S.Rangasamy College of Technology, Tiruchengode, B.E in Artificial Intelligence and Machine Learning 2022 – 2026

- CGPA : 8.38
- Relevant Coursework: Object Oriented Programming, Databases, Data Structures and Algorithms, Operating Systems, Computer Networks.

Technical Skills

Languages: Java, Python, SQL, Kotlin.

Technologies/Tools: Machine Learning, Large Language Models (LLMs), PyTorch, Generative AI, Agentic AI, Computer Vision, Scikit-learn, TensorFlow, MediaPipe, NumPy, Pandas, Matplotlib , Seaborn, Flask, REST APIs, Android Studio, PyAudio , SpeechRecognition, RPi.GPIO, GIT, Fusion 360 , AutoCAD, RPLIDAR SDK, MySQL, MongoDB

Experience

Backend Developer Intern, MetatronCube – Canada Jan 2025 – Apr 2025

- **Tech Stack:** Python, OpenCV, NumPy, Matplotlib, RPLIDAR SDK, RPLIDAR A1, RANSAC, Computational Geometry, Point Cloud Processing
- Developed a real-time indoor mapping system using 2D RPLIDAR by converting polar coordinates into accurate Cartesian layouts, implementing geometry-based wall detection, RANSAC line fitting, and corner detection for precise room dimension estimation.
- Engineered a modular spatial data processing pipeline with Python, NumPy, and OpenCV for point cloud preprocessing, noise filtering, feature extraction, and real-time floor plan generation, enabling applications in indoor navigation, robotics, and smart architecture.

Projects

Safeguard AI Link

- **Tech Stack:** Python, TensorFlow, OpenCV, MediaPipe, DeepFace, Flask, Flask-SocketIO, MongoDB, Computer Vision, CNN
- Designed and developed an AI-powered intelligent surveillance system leveraging Computer Vision, Deep Learning, TensorFlow, OpenCV, MediaPipe, and DeepFace for real-time gesture recognition, facial emotion analysis, and automated emergency alert generation.
- Engineered a low-latency real-time inference pipeline using Flask, Flask-SocketIO, and MongoDB to enable live video streaming, multimodal AI decision-making, event management, and scalable deployment for smart surveillance applications.

T2T PhonoKit Link

- **Tech Stack:** Python, Flask, SpeechRecognition, Vosk, Raspberry Pi, SQLite/MySQL, REST API, Natural Language Processing (NLP), Speech Processing, Wake Word Detection, Real-Time Audio Processing
- Developed an AI-powered voice-assisted learning system for dyslexic students using Speech Recognition, Natural Language Processing (NLP), and wake-word detection to evaluate pronunciation accuracy against teacher-assigned words in real time.

- Engineered a Raspberry Pi-based embedded AI solution integrating Vosk/KaldiRecognizer, Flask, gTTS, and a database backend for real-time speech processing, progress tracking, and scalable web-based educational content management.

LLM Model Selection & Optimization Link

- **Tech Stack:** Python, Ollama, NVIDIA NIM API, LangChain, Hugging Face Transformers, Gemma 4, Qwen 3, Embedding Models, REST API, VS Code
- Evaluated and compared locally deployed Large Language Models (LLMs), including **Gemma 3**, **Qwen 3**, based on inference speed, response quality, context window, resource utilization, and deployment feasibility to identify the most suitable model for AI assistant applications.
- Developed a modular inference pipeline integrating locally hosted LLMs through Ollama and NVIDIA NIM embedding APIs, enabling efficient model benchmarking, API-based inference, and a scalable foundation for future RAG and AI assistant development.

Area of Interest

Passionate about Problem Solving and continuously sharpening my skills in Data Structures and Algorithms (DSA). I enjoy breaking down complex problems into elegant, efficient solutions and exploring algorithmic thinking through real-world coding challenges.

Regularly practice on competitive programming platforms and actively follow coding patterns, optimization techniques, and system design fundamentals to stay industry-ready.

Achievements

- Earned a Python Certificate from HackerRank — (sthiyaneshwar94)
- Achieved 100+ problems solved badge on LeetCode — (thiyaneshwars)
- Finalist at Entropy Hackathon, VIT, selected among top teams for innovative AI-based solutions.
- Participated and recognized in national competitions like Bhumi Hackathon and Speak for India.
- Represented at the Karate State Level Championship, displaying strong discipline and competitive spirit.
- Successfully conducted a workshop on Artificial Intelligence at NTTF Trichy.
- Successfully earned the Oracle Cloud Infrastructure 2025 DevOps Professional Certification..